



Sited, Not Chosen

Interviewing Cyclists at the Bike-Rack Bank the Horizon Register Predicted and the One They Actually Fill

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Abstract. The School of Emergent Priorities' Horizon Register flags precinct-level growth in badge-registered bike-commute sign-ups as a leading indicator for siting new cycle infrastructure. A rising trend in the West Wing precinct led Facilities to install a 40-slot covered bike-rack bank outside the Wing's north entrance ahead of the term in which demand was expected to arrive. We report a mixed-methods evaluation of that siting call across the following 12-week term: fixed-time occupancy counts at the forecast-sited bank and at an existing, unforecast 24-slot bank outside the Library, and semi-structured interviews with 42 cyclists parking at either location, thematically coded for their stated reason. The forecast-sited bank averaged 22% occupancy against the unforecast bank's 115% (overflow bikes counted against nominal capacity); the Register's precinct-demand index correlates weakly with realised weekly occupancy ($r = 0.34$), and removing its trend-extrapolation term changes this negligibly ($r = 0.32$, n.s.), which we retain for completeness. The reason cyclists most often gave — unprompted proximity to the building they were actually visiting — correlates far more strongly with occupancy ($r = 0.79$) than any input the Register's model consumes. We discuss what a badge-swipe leading indicator may be measuring, if not incipient demand for the infrastructure it is used to site.

Introduction

Anticipatory infrastructure planning promises to convert a reactive build-to-complaint cycle into one that provisions ahead of need, forecasting where demand will land from an early behavioural signal rather than waiting for an existing facility to visibly overflow. For cycling infrastructure specifically, the evidence that new provision can generate its own uptake is reasonably strong: a new segregated lane can measurably lift cycling traffic within a year of opening [1], and gender-disaggregated natural experiments find comparable gains concentrated among previously under-served riders [2]. What this literature does not establish is whether a forecast used to decide **where** that provision goes, ahead of any observed use of the specific site, predicts the location cyclists will actually choose — as opposed to predicting,

more modestly, where an administrative signal happens to be easiest to collect.

At Slop University, the School of Emergent Priorities' Horizon Register maintains a precinct-level demand index from badge-registered sign-ups to the University's bike-commute scheme, smoothed into a leading indicator that Facilities consults when a new rack bank is due. A rising index reading for the West Wing precinct saw a 40-slot covered bank installed outside the Wing's north entrance in time for the term in which the Register's model expected demand to materialise. We ask whether the bank the Register sited is the bank cyclists actually use, and, if not, what the index the Register consults is tracking instead.

We make three contributions, each hedged to the evidence available. First, we assemble a term-long occupancy record at the forecast-sited bank and at an existing, unforecast bank,

independent of the Register’s own precinct reading. Second, we interview 42 cyclists at both locations and thematically code their stated reason for parking there, recovering a small taxonomy of motivations the Register’s model does not include as inputs. Third, we show that a simple post-hoc geographic measure — walking distance to the cyclist’s actual destination — correlates with realised occupancy far more strongly than the Register’s own precinct-demand index, and suggest this discrepancy may generalise to other badge-derived leading indicators used for siting decisions elsewhere in the University.

Related work

Bicycle-parking research has moved from counting occupied racks to modelling why a cyclist chooses one facility over another: stated-preference work finds cyclists strongly value short walking distances to their actual destination and a preference for covered over uncovered racks, with a measurable willingness to pay for both [3], and combining stated preference with count data materially improves demand models once walking distance sensitivity is properly weighted [4]. A systematic review of the wider parking-behaviour literature finds the same distance and facility-type factors recurring across settings [5]. Facility choice models built to forecast bike-lane usage before construction carry a standing caution of their own: added infrastructure does not guarantee cyclists switch to using it [6], a finding echoed by frameworks that fuse subjective rider perception with objective site measures precisely because either alone under-predicts actual use [7], [8].

A separate literature treats a demand forecast as capable of shaping the outcome it predicts rather than merely describing it: forecasting has been characterised as a self-fulfilling exercise wherever future supply decisions and future demand are coupled through the same feedback loop [9], a dynamic documented in long-run transport-capacity modelling [10], in AI-assisted economic forecasting [11], and experimentally, where a forecast of change measurably shifts behaviour before any change is realised [12]. A comparable predicted-versus-actual gap is well documented outside transport entirely: non-domestic buildings’ modelled energy performance routinely diverges from metered use once occu-

pants’ actual behaviour is measured against the model’s assumptions [13]. Prior Slop University work has found a structurally similar pattern in the Horizon Register’s own operation — its early-warning alerts predict which flagged priorities are subsequently resourced far better from a unit’s reporting volume than from the priority’s assessed novelty [14] — and a related instrument elsewhere in the University credits a forecast with a lead time that mostly evaporates once scored against an independent ground truth rather than the forecast’s own predicted date [15]. We extend this line of inquiry from the Register’s alerts to its siting recommendations, and add a qualitative account, coded by the six-phase method of [16], of what cyclists say they are actually optimising for.

Method

The West Wing bank (40 covered slots, installed at the start of the study term) and the Library bank (24 uncovered slots, in place for at least three prior years) were each observed at three fixed times per week — 09:00, 13:00, and 16:30, chosen to span the University’s core teaching hours — for the 12-week term, yielding 36 occupancy counts per site. A bicycle locked to a railing or sign post within five metres of the Library bank once its slots were full was counted as overflow against the bank’s nominal capacity, so a site’s occupancy rate can exceed 100%.

Site	Slots	Counts	Interviewed
West Wing (forecast-sited)	40	36	18
Library (un-forecast)	24	36	24

Cyclists arriving at either bank during a count window were invited to a short semi-structured interview (median 9 minutes) asking why they had chosen that bank, with follow-up prompts on cover, distance, and prior routine. Interviews continued at each site until two consecutive sessions of three interviews introduced no new code, consistent with the saturation point reported for comparably scoped qualitative studies [17]; 42 interviews were completed in total (18 West Wing, 24 Library). Transcripts were coded using the six-phase thematic-analysis procedure of [16] by both authors independently, with disagreements resolved by discussion; codes were not mutually exclusive, so an

interview could contribute to more than one category.

The Horizon Register’s precinct-demand index updates weekly from badge-swipe counts S_t recorded at the University’s bike-commute scheme kiosks in the West Wing precinct,

$$I_t = \alpha S_t + (1 - \alpha)I_{t-1}, \quad \alpha = 0.4,$$

the School’s standing smoothing constant for precinct-level indices. A bank is recommended for siting the first term in which I_t clears a fixed threshold calibrated from the Register’s founding cohort of precincts. We correlate I_t ’s twelve weekly readings for the West Wing precinct against that bank’s realised weekly occupancy, and repeat the correlation with the trend term removed ($I_t := S_t$, no smoothing) as an ablation isolating its contribution. As a post-hoc comparison outside the Register’s own model, we also compute each week’s mean walking distance from parked bikes to the building entrance interviewees named as their destination, correlated against the same occupancy series. All three correlations are bootstrapped by leave-one-week-out resampling (15 resamples each) to report a spread rather than a single point estimate.

Results

Figure 1 reports weekly occupancy at both banks. The West Wing bank held between 15% and 30% occupancy throughout the term (mean 22%); the Library bank held between 95% and 140% (mean 115%), with overflow bikes present in every count window from week 4 onward. The two series never cross.

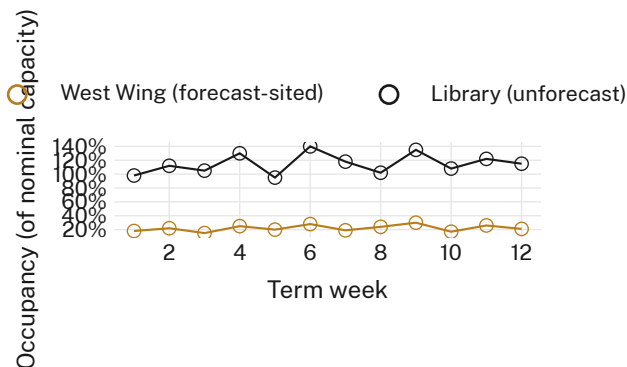


Figure 1: Weekly occupancy at the forecast-sited West Wing bank and the unforecast Library bank: the two series hold their relative positions across all 12 weeks of the term, with no week in which the sited bank out-occupies the unforecast one.

Figure 2 summarises the coded interview reasons. Proximity to an interviewee’s actual destination was the most-cited reason at both sites (14 of 18 West Wing interviewees, 21 of 24 Library interviewees). Prior habit — routing to a bank the cyclist had used before the study term began — was cited by 16 of 24 Library interviewees against 2 of 18 at the newly installed West Wing bank, where such a habit could not yet exist. Visibility and a sense of theft deterrence favoured the busier Library bank (11 of 24 against 3 of 18); the covered West Wing bank was the only site where shelter from weather was volunteered as a reason, by 6 of 18 interviewees. Separately from the coded categories, 9 of 24 Library interviewees said, unprompted, that they had not known the West Wing bank existed.

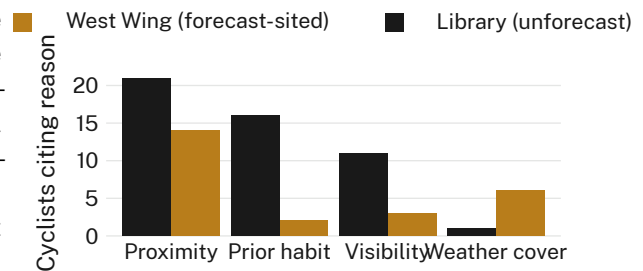


Figure 2: Coded reasons cyclists gave for parking where they did, by site: proximity to the interviewee’s own destination dominates at both banks, while prior habit and perceived visibility favour the established, unforecast bank.

Figure 3 reports the bootstrapped correlations between realised weekly occupancy and the three candidate predictors. The Register’s full precinct-demand index correlates weakly with occupancy (median $r = 0.34$ across resamples); removing its trend-extrapolation term leaves this essentially unchanged (median $r = 0.32$, overlapping interquartile ranges), which we retain for completeness. The post-hoc distance-to-entrance measure — built from the reason interviewees themselves gave most often, and consuming no input the Register’s model sees — correlates far more strongly and consistently (median $r = 0.79$, narrower spread).

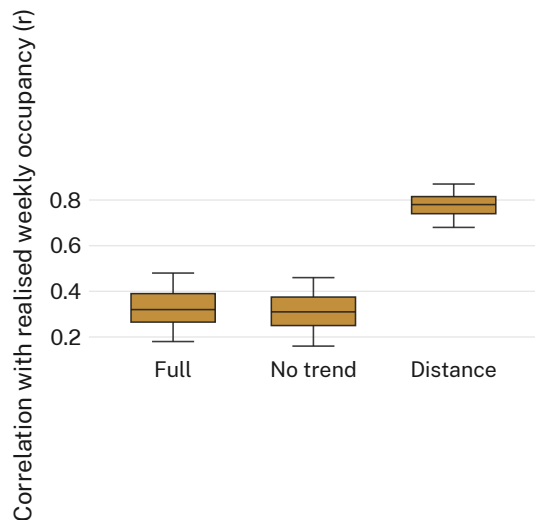


Figure 3: Bootstrapped correlation with realised weekly occupancy: removing the Register model’s trend-extrapolation term changes its already-weak correlation negligibly; a post-hoc distance-to-entrance measure the model never consumes correlates roughly twice as strongly.

Discussion

These results suggest the Horizon Register’s precinct-demand index tracks something closer to a precinct’s badge-swipe reporting activity than to where cyclists will actually leave their bikes, consistent with this University’s own prior finding that a related Register alert predicts a unit’s reporting volume more reliably than the condition it is meant to detect [14]. The finding does not indicate the index is miscalibrated for the purpose Facilities used it for; siting decisions must be made before a new building’s occupants have established any parking routine at all, and a precinct-level badge count is at least available at that point, where a destination-level distance measure is not. Rather, the gap between the two correlations may indicate that “anticipatory” siting substitutes an administratively convenient proxy for the behavioural variable it is intended to anticipate, a substitution the wider forecasting literature suggests is easy to make and hard to notice once a scheme is operating [9], [10]. We do not read the low proximity of the West Wing bank to its own precinct’s index as evidence that covered parking is unwanted: six interviewees who did use it named the cover unprompted, suggesting demand for it may simply be concentrated among riders whose destination happens to sit nearby, rather than distributed evenly across the precinct the index measures.

Limitations

Our sample of two banks and one term is modest, though the consistency of the occupancy gap across all 12 weeks, and of the coded reasons across both authors’ independent passes, suggests the pattern is not an artefact of either site’s particular week or a single coder’s judgement. Relying on interviewees’ self-reported destination to construct the distance measure could overstate its correlation with occupancy if cyclists rationalise their parking choice after the fact rather than reporting the reason that drove it; if anything, this would work against the finding by inflating agreement between a stated and an enacted preference that a purely behavioural measure would not assume. The single precinct and single term studied precludes claims about whether the Register’s index performs better once a precinct’s occupants have settled into a routine, a question we consider a natural extension of the present, already consistent, findings.

Conclusion

We evaluated a Horizon Register siting recommendation against a term of occupancy counts and cyclist interviews at the bank it sited and the bank cyclists actually filled, finding the Register’s precinct-demand index a weak predictor of realised use and a simple, unconsumed distance measure a substantially stronger one. An ablation removing the index’s trend term left this picture unchanged. We suggest siting recommendations for cycle infrastructure carry a destination-distance estimate alongside any precinct-level demand index as standard practice. Future work should track the West Wing bank across a full year to test whether occupancy converges on the Register’s forecast as routines settle, and should ask whether the same proxy substitution recurs in the Register’s other siting recommendations across the University.

References

- [1] P. Rérat and A. Schmassmann, “Build it and they will come? The effects of a new infrastructure on cycling practices and experiences,” *Transportation Research Interdisciplinary Perspectives*, 2024, doi: [10.1016/j.trip.2024.101121](https://doi.org/10.1016/j.trip.2024.101121).
- [2] M. Pellicer-Chenoll, L. Antón-González, I. Villarrasa-Sapiña, J. Devís-Devís, L.-M.

- González, and M. Pans, “Effects of building cycling infrastructure on bicycle use: Differences by gender through a longitudinal natural experiment study,” *Research in Transportation Economics*, 2025, doi: [10.1016/j.retrec.2025.101531](https://doi.org/10.1016/j.retrec.2025.101531).
- [3] D. Kohlrantz and T. Kuhnimhof, “Cyclists’ heterogeneous parking preferences and their implications for bicycle parking facilities,” *Transportation Research Part A: Policy and Practice*, 2025, doi: [10.1016/j.tra.2024.104298](https://doi.org/10.1016/j.tra.2024.104298).
- [4] D. Kohlrantz and T. Kuhnimhof, “Planning for bicycle parking: Predicting demand using stated preference and count data,” *Journal of Cycling and Micro-mobility Research*, 2024, doi: [10.1016/j.jcmr.2024.100011](https://doi.org/10.1016/j.jcmr.2024.100011).
- [5] E. Heinen and R. Buehler, “Bicycle parking: a systematic review of scientific literature on parking behaviour, parking preferences, and their influence on cycling and travel behaviour,” *Transport Reviews*, 2019, doi: [10.1080/01441647.2019.1590477](https://doi.org/10.1080/01441647.2019.1590477).
- [6] A. Kojima and H. Kubota, “Modeling cyclists’ facility choice and its application in bike lane usage forecasting,” *IATSS Research*, 2018, doi: [10.1016/j.iatssr.2017.06.006](https://doi.org/10.1016/j.iatssr.2017.06.006).
- [7] K. Tian, Y. Zheng, Z. Sun, Z. Yin, K. Zhu, and C. Liu, “A Comprehensive Framework for Evaluating Cycling Infrastructure: Fusing Subjective Perceptions with Objective Data,” *Sensors*, 2025, doi: [10.3390/s25041168](https://doi.org/10.3390/s25041168).
- [8] T. Götschi, A. Castro, M. Deforth, L. Miranda-Moreno, and S. Zangenehpour, “Towards a comprehensive safety evaluation of cycling infrastructure including objective and subjective measures,” *Journal of Transport & Health*, 2018, doi: [10.1016/j.jth.2017.12.003](https://doi.org/10.1016/j.jth.2017.12.003).
- [9] O. C. King and M. Mertens, “Self-fulfilling Prophecy in Practical and Automated Prediction,” *Ethical Theory and Moral Practice*, 2023, doi: [10.1007/s10677-022-10359-9](https://doi.org/10.1007/s10677-022-10359-9).
- [10] S. P. Blainey and J. M. Preston, “Predict or prophesy? Issues and trade-offs in modelling long-term transport infrastructure demand and capacity,” *Transport Policy*, 2019, doi: [10.1016/j.tranpol.2018.12.001](https://doi.org/10.1016/j.tranpol.2018.12.001).
- [11] E. Beretta, V. Dianova, G. Pankratz, and U. Schollmeyer, “AI-Based Forecasting and Market Expectations: A Self-Fulfilling Prophecy?,” *Review of Political Economy*, 2025, doi: [10.1080/09538259.2025.2562206](https://doi.org/10.1080/09538259.2025.2562206).
- [12] D. Petropoulos Petalas, H. van Schie, and P. Hendriks Vettehen, “Forecasted economic change and the self-fulfilling prophecy in economic decision-making,” *PLOS ONE*, 2017, doi: [10.1371/journal.pone.0174353](https://doi.org/10.1371/journal.pone.0174353).
- [13] A. C. Menezes, A. Cripps, D. Bouchlaghem, and R. Buswell, “Predicted vs. actual energy performance of non-domestic buildings: Using post-occupancy evaluation data to reduce the performance gap,” *Applied Energy*, 2012, doi: [10.1016/j.apenergy.2011.11.075](https://doi.org/10.1016/j.apenergy.2011.11.075).
- [14] V. Marris and P. Umbile, “Flagged, Then Funded: Validating the Horizon Register’s Early-Warning Alerts Against What a Priority Actually Receives,” *Slopp University technical report*, 2026, doi: [10.5555/slop.4wzjf0](https://doi.org/10.5555/slop.4wzjf0).
- [15] F. Okoro, K. Lindqvist, and C. Beng, “Anticipated, Not Averted: A Field Validation of a Predictive Stationery-Cupboard Reorder Forecast Against a Term of Logged Stock-Outs,” *Slopp University technical report*, 2026, doi: [10.5555/slop.plhjq7](https://doi.org/10.5555/slop.plhjq7).
- [16] V. Braun and V. Clarke, “Using thematic analysis in psychology,” *Qualitative Research in Psychology*, 2006, doi: [10.1191/1478088706qp0630a](https://doi.org/10.1191/1478088706qp0630a).
- [17] G. Guest, A. Bunce, and L. Johnson, “How Many Interviews Are Enough? An Experiment with Data Saturation and Variability,” *Field Methods*, 2006, doi: [10.1177/1525822X05279903](https://doi.org/10.1177/1525822X05279903).